Solution:

Year	Income	3-Year semi-total	Semi-average
2008	46.17	161.63	53.877
2009	51.65		
2010	63.81		
2011	70.99	247.54	82.513
2012	84.91		
2013	91.64		

Difference between the central years = 2012 – 2009 = 3

Difference between the semi-averages = 82.513 – 53.877 = 28.636

Increase in trend value for one year = $\frac{28.636}{3} = 9.545$

Trend values for the previous and successive years of the central years can be calculated by subtracting and adding respectively, the increase in annual trend value.

Example 7.3

Population of India for 7 successive census years are given below. Find the trend values using semi-averages method.

Census Year	1951	1961	1971	1981	1991	2001	2011
Population (in lakhs)	301.2	336.9	412.0	484.1	558.6	624.1	721.4

Solution:

Trend values using semi average method

Census Year	Population (in lakhs)	3-year semi-total	3-year semi-average	Trend values
1951	301.2	1050.1	350.03	278.86
1961	336.9			350.03
1971	412.0			421.2
1981	484.1			492.37
1991	558.6	1904.1	634.7	563.54
2001	624.1			634.71
2011	721.4			705.88

Difference between the years = 2001 – 1961 = 40

Difference between the semi-averages = 634.7 – 350.03 = 284.67

Increase in trend value for 10 year = $\frac{284.67}{4} = 71.17$

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